Area and surface area

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 9.2 × 10

2. 900 ÷ 100

3. Double 9.5

4. Half of 54

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

7. Solve x + 9 = 23.

8. Find 0.8 × 0.7. Is the product below both factors?

9. Find $\frac{3}{4} \times \frac{4}{5}$.

10. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

11. 5 × 2

12. A box is 5 cm by 2 cm by 3 cm. Find its volume.

13. A rectangle has whole-number side lengths, perimeter 24 cm, and area 35 square cm. Find its side lengths.

14. A rectangle has length 10 cm and width 3 cm. Find area and perimeter.

15. A parallelogram has base 8 cm, height 3 cm, and slanted side 5 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 16 cm and 3 cm. Find area using either leg as base.

17. A triangle has area 24 square cm and base 16 cm. Find its matching height.

18. A11 cm by 5 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

19. A triangle and parallelogram share base 8 and height 3. Find each area and their ratio.

20. A closed box measures 2 cm by 3 cm by 8 cm. Find surface area.

21. A cube has side 3 cm. Find surface area and volume; label units.

22. A square pyramid has base side 6 cm and triangular face height 8 cm. Find surface area.

23. A prism has 3-4-5 right-triangle ends and length 8 cm. Find its surface area.

CHALLENGE / Explain your plan and check the result

C1. An L-shaped floor comes from a10 m by 8 m rectangle with a4 m by 3 m corner removed. Tiles cover 2 square m each. Tiles are sold in packs of 6. How many packs must be bought, and how much area will the unused tiles cover?

C2. A tent is a triangular prism 8 m long with right-triangle ends of sides 3, 4, 5 m. The 3 m side rests on the ground. Find the fabric area for both ends and the two other rectangular sides, with no floor. Compare with a fully closed tent.